

1. Input

Section, materials and resistance basis exactly as analysed.

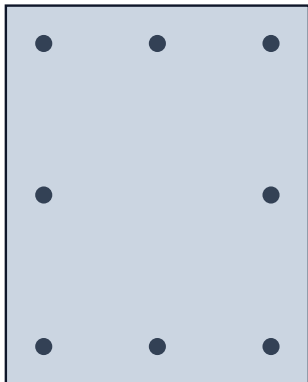
Identity

Project	ACI-EB 01 - Rectangle 400 x 500 m.	Section	Benchmark 01 - Rectangle 400 x 50.
Calculation profile	ACI 318-19(22) — Whitney equivalence.	Mechanics	Equivalent rectangular stress block
Resistance profile	ACI CODE-318-19 (Reapproved 202.	Profile edition	2019 (Reapproved 2022)
Verification status	draft	Concrete model basis	Code-default
Method id	aci-318-19-22-global-strength-reduct.	Units	mm · N · MPa; forces reported in kN,.
Sign convention	Compression positive; $M_x = \Sigma F \cdot (y - y_c)$.		

Section

Cross-section (mm, about the net concrete centroid)

8 bars



Section properties

Gross concrete area A_g	200,000 mm ²
Reinforcement area A_s	3,200 mm ²
Reinforcement ratio ρ	1.600 %
Bar count	8
Net centroid	(0.0, 0.0) mm
Bounding box	400.0 × 500.0 mm
Solids / holes	1 / 0
Perimeter	1,800.0 mm

Materials

Name	Concrete C40	Standard	ACI318
f_{ck}	40.0 MPa	Model	aci-whitney-block
ϵ_{cu}	0.00300	α	0.8500
E_c	29,725 MPa	Tension	ignored
Name	Rebar fy 420	Standard	ACI318
f_y	420.0 MPa	E_s	200,000 MPa
Model	elastic-perfectly-plastic	ϵ_y	0.002100
ϵ_u	not declared	Grades used	1

Design resistance basis

Format	Global resultant strength-reduction f.	Transverse reinforcement	other
ϕ compression (ties)	0.650	ϕ compression (spiral)	0.750
ϕ tension	0.900	ϵ_t , limit	0.00510
Transition rule	yield-plus-strain	Axial cap	0.800 × factored compression pole

Sampling and numerical evidence

Governing adaptive points	4825	Adaptive directions	128
Adaptive stations	48	Fixed display points	597
Fixed display directions	36	Fixed stations	22
Direction tolerance	0.750 %	Worst interpolation error	0.7122 %
Tolerance reached	yes	Concrete discretization	exact polygon clipping
Strain domain	concrete-pivot-ultimate		

Scope and limitations

This document reports a preview calculation. It is not an accepted design result and must not be released as a design report.

Resistance is reported in two stages: Nominal is the reference before the resistance treatment, Design is the usable resistance. Only the Design surface is compared with factored ULS demand.

Concrete compression is the code-equivalent rectangular block over $a = \beta_1 \cdot c$, obtained by exact polygon clipping. There is no concrete integration mesh, and the stress between a and c is zero.

Overview interaction diagrams are geometric plane cuts of the fixed 22-station, 10° direction grid. Adaptive design points are reserved for

governing resistance checks and exact demand meridians.

Warning: Equivalent-block neutral-axis refinement did not reach its requested tolerance.

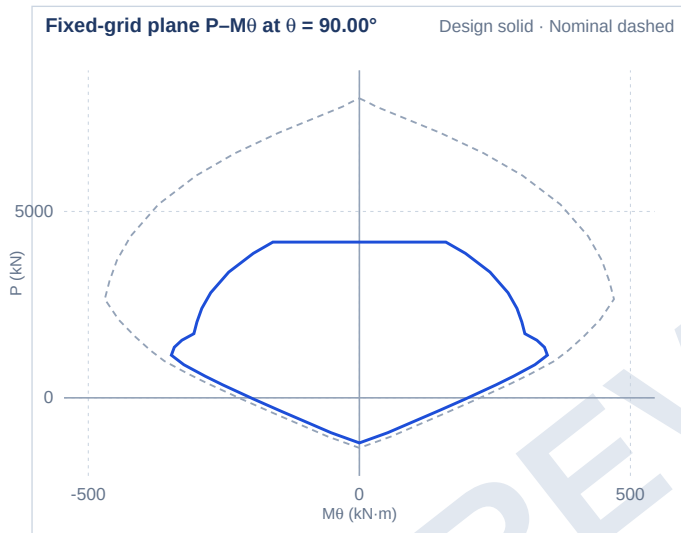
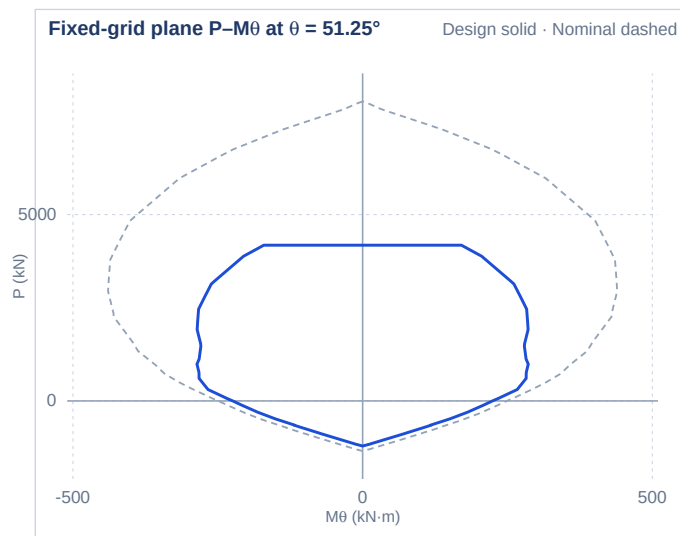
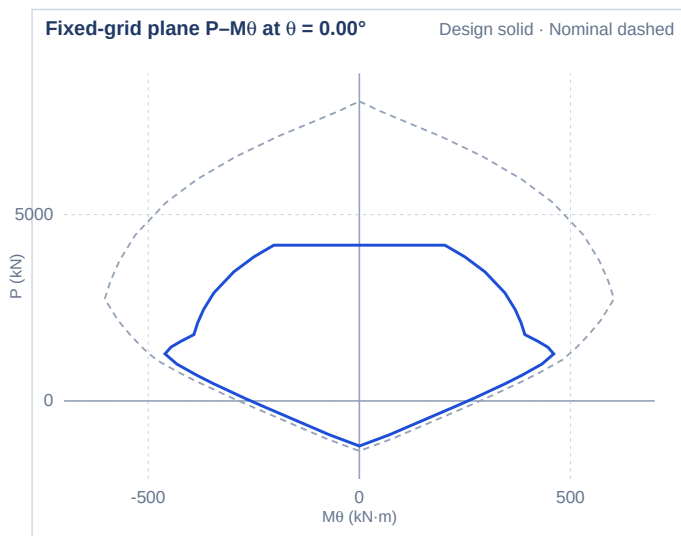
Reinforcement schedule

#	Bar id	(mm)	X (mm)	Y (mm)	As (mm ²)	Steel
1	1	22.6	-150.0	-200.0	400.0	Rebar fy 420
2	2	22.6	0.0	-200.0	400.0	Rebar fy 420
3	3	22.6	150.0	-200.0	400.0	Rebar fy 420
4	4	22.6	150.0	0.0	400.0	Rebar fy 420
5	5	22.6	150.0	200.0	400.0	Rebar fy 420
6	6	22.6	0.0	200.0	400.0	Rebar fy 420
7	7	22.6	-150.0	200.0	400.0	Rebar fy 420
8	8	22.6	-150.0	0.0	400.0	Rebar fy 420

2. Section capacity

Nominal reference and Design resistance. Only Design is compared with factored demand.

Maximum design axial resistance applied at 4178.3 kN; it appears as the flat top of each diagram.



3. Factored ULS demand

Governing check is the proportional 3D ray against the Design surface.

#	Combination	P _u (kN)	M _{ux} (kN·m)	M _{uy} (kN·m)	θ _L (°)	φ	Class	UR	Verdict
1	Audit ray 55% - NA 26.93 d.	661.2	97.5	121.5	51.3	0.694	transition	0.550	ADEQUATE
2	Cardinal audit 85% - NA 0 d.	2178.8	0.0	242.0	90.0	0.650	compression-co.	0.850	ADEQUATE
3	Cardinal audit 85% - NA 90 .	1606.2	330.7	0.0	0.0	0.650	compression-co.	0.850	ADEQUATE

UR is the governing proportional utilization: the factored demand vector scaled until it meets the Design surface. A fixed-axial ratio is reported per combination in its detail section and is a secondary diagnostic only.

4.1 Audit ray 55% - NA 26.93 deg, c/D 0.55

$P_u = 661.2 \text{ kN}$ · $M_{ux} = 97.5 \text{ kN}\cdot\text{m}$ · $M_{uy} = 121.5 \text{ kN}\cdot\text{m}$ · $\theta_L = 51.25^\circ$

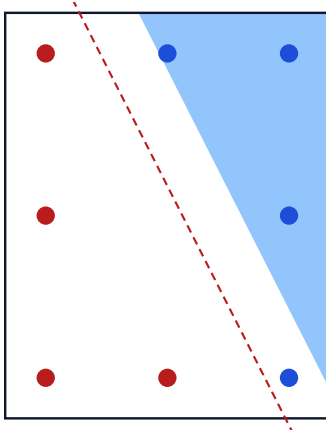
ADEQUATE · utilization 0.5500 · fixed-P diagnostic 0.5491

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

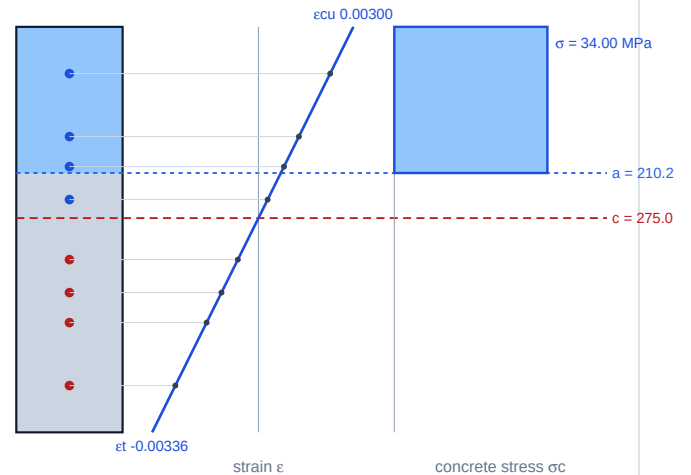
Transverse section — neutral axis and compression zone

a=210



Longitudinal section — strain and stress over the depth

Dθ=583

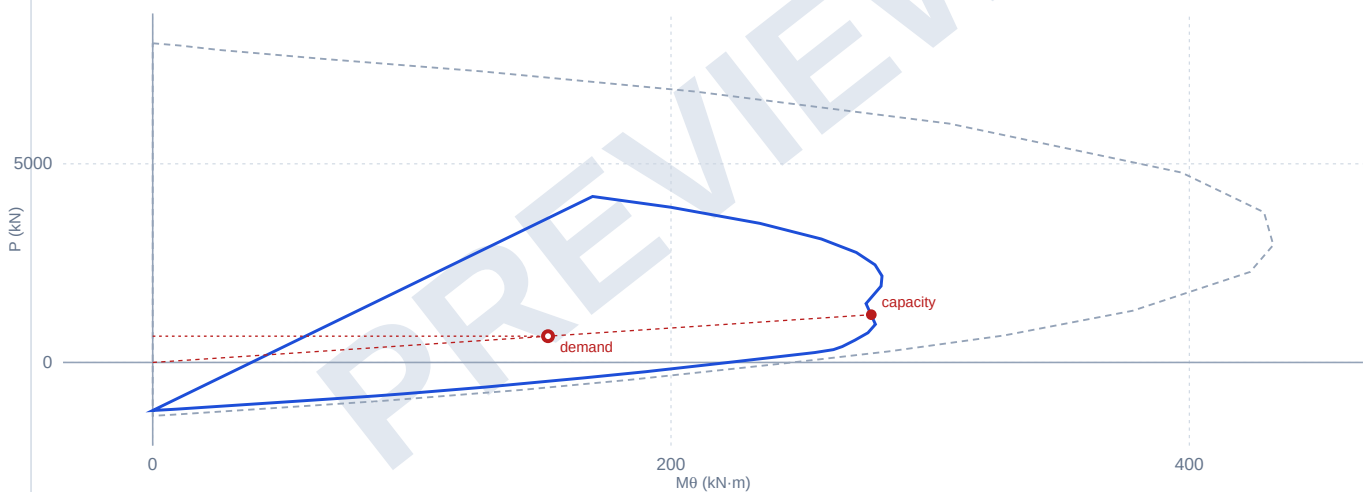


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P-Mβ at βeq = 63.071°, with the load point

solid = adaptive Design · dashed = fixed Nominal · points are projected on βeq



Concrete and resultants

Block area $A_{c,blk}$	54,702.0 mm ²	Block centroid	(121.4, 95.3) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	1,859.9 kN
Concrete M_{cx}	177.3 kN·m	Concrete M_{cy}	225.8 kN·m
Nominal P_n	1,732.5 kN	Nominal M_{nx}	255.6 kN·m
Nominal M_{ny}	318.5 kN·m	ϕ	0.6940
Design ϕP_n	1,202.3 kN	Design ϕM_{nx}	177.4 kN·m
Design ϕM_{ny}	221.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	515.8	-2.627e-3	-420.00	-168.00	no
2	0.0	-200.0	400.0	382.1	-1.168e-3	-233.70	-93.48	no
3	150.0	-200.0	400.0	248.4	2.904e-4	58.09	23.23	no
4	150.0	0.0	400.0	157.8	1.279e-3	255.71	88.68	yes

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	67.2	2.267e-3	420.00	154.40	yes
6	0.0	200.0	400.0	201.0	8.077e-4	161.55	51.02	yes
7	-150.0	200.0	400.0	334.7	-6.512e-4	-130.24	-52.10	no
8	-150.0	0.0	400.0	425.3	-1.639e-3	-327.86	-131.15	no

Solver and admissibility evidence

Neutral-axis angle θ	26.9290°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β1	0.7643
Projected depth Dθ	583.07 mm	Controlling εt	0.002627
Controlling bar	1	Classification	transition
Axial cap governs	no	Component residual	1.16e-10
Strain admissibility	admissible	Governing utilization	0.5500

4.2 Cardinal audit 85% - NA 0 deg, c/D 0.55

$P_u = 2178.8 \text{ kN}$ · $M_{ux} = 0.0 \text{ kN}\cdot\text{m}$ · $M_{uy} = 242.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 90.00^\circ$

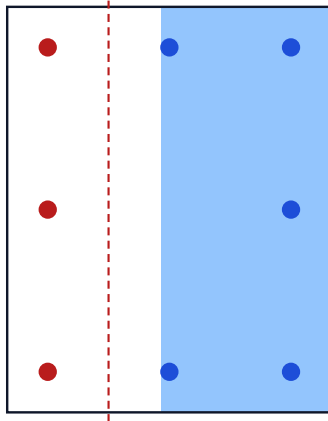
ADEQUATE · utilization 0.8500 · fixed-P diagnostic 0.8163

Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

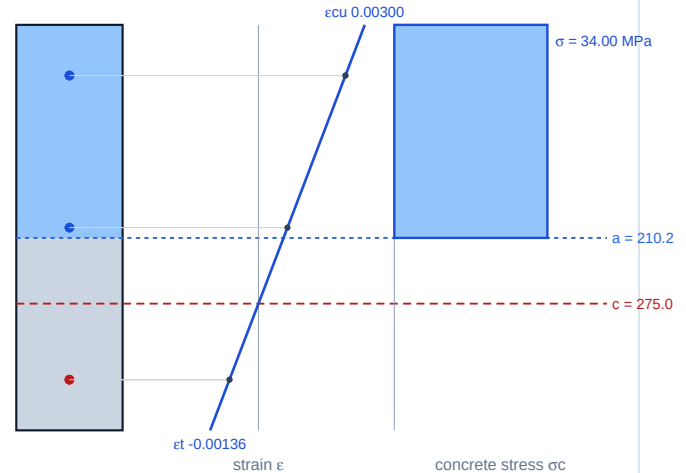
Transverse section — neutral axis and compression zone

a=210



Longitudinal section — strain and stress over the depth

Dθ=400

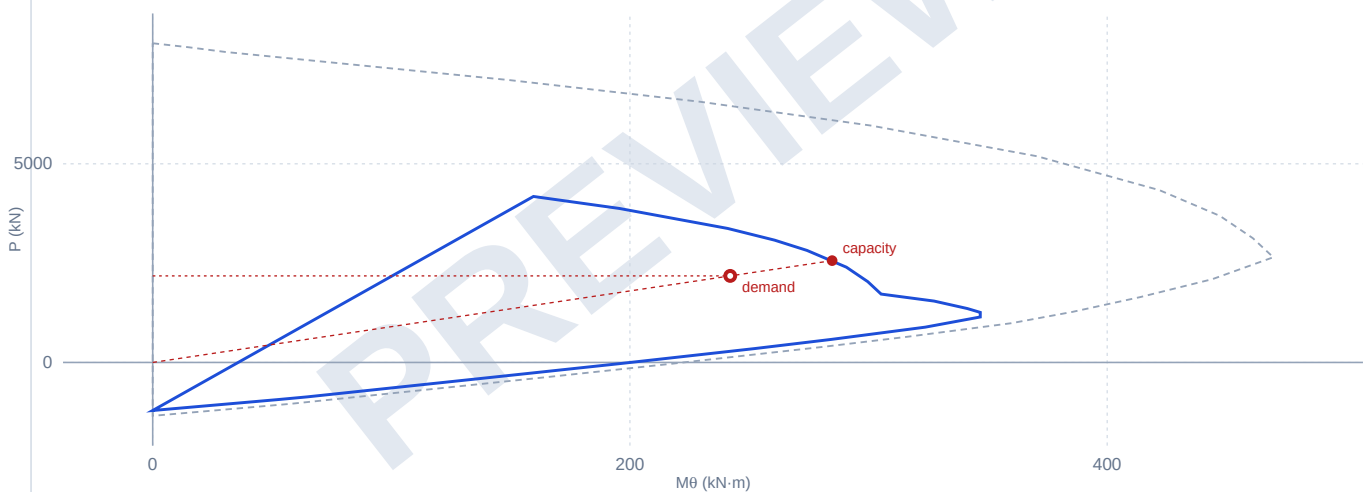


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct P-Mβ at βeq = 90.000°, with the load point

solid = adaptive Design · dashed = fixed Nominal · points are projected on βeq



Concrete and resultants

Block area $A_{c,blk}$	105,089.3 mm ²	Block centroid	(94.9, -0.0) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	3,573.0 kN
Concrete M_{cx}	-0.0 kN·m	Concrete M_{cy}	339.1 kN·m
Nominal P_n	3,943.6 kN	Nominal M_{nx}	-0.0 kN·m
Nominal M_{ny}	438.1 kN·m	ϕ	0.6500
Design ϕP_n	2,563.3 kN	Design ϕM_{nx}	-0.0 kN·m
Design ϕM_{ny}	284.7 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
2	0.0	-200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
3	150.0	-200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
4	150.0	0.0	400.0	50.0	2.455e-3	420.00	154.40	yes

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	200.0	8.182e-4	163.64	51.85	yes
7	-150.0	200.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no
8	-150.0	0.0	400.0	350.0	-8.182e-4	-163.64	-65.45	no

Solver and admissibility evidence

Neutral-axis angle θ	0.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β1	0.7643
Projected depth Dθ	400.00 mm	Controlling εt	0.000818
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	-1.75e-10
Strain admissibility	admissible	Governing utilization	0.8500

4.3 Cardinal audit 85% - NA 90 deg, c/D 0.55

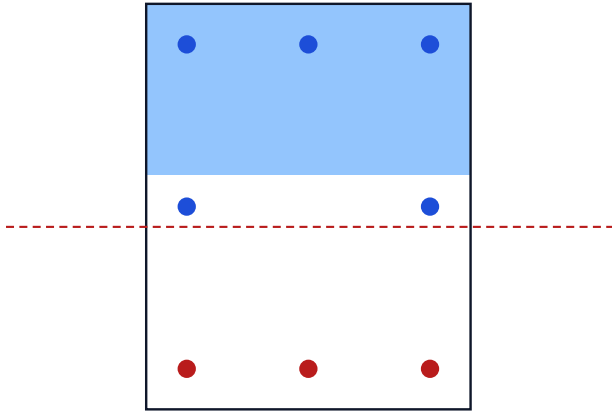
$P_u = 1606.2 \text{ kN}$ · $M_{ux} = 330.7 \text{ kN}\cdot\text{m}$ · $M_{uy} = 0.0 \text{ kN}\cdot\text{m}$ · $\theta_L = 0.00^\circ$

ADEQUATE · utilization **0.8500** · fixed-P diagnostic **0.7840**

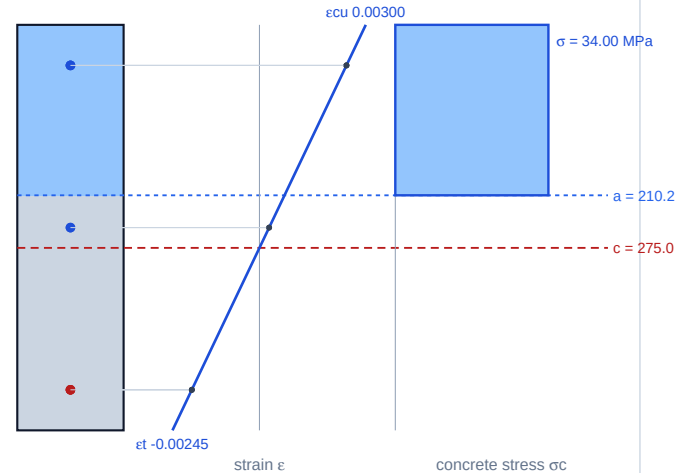
Factored ULS demand checked by the equivalent-block proportional-ray solver.

Section state

Transverse section — neutral axis and compression zone $a=210$



Longitudinal section — strain and stress over the depth $D\theta=500$

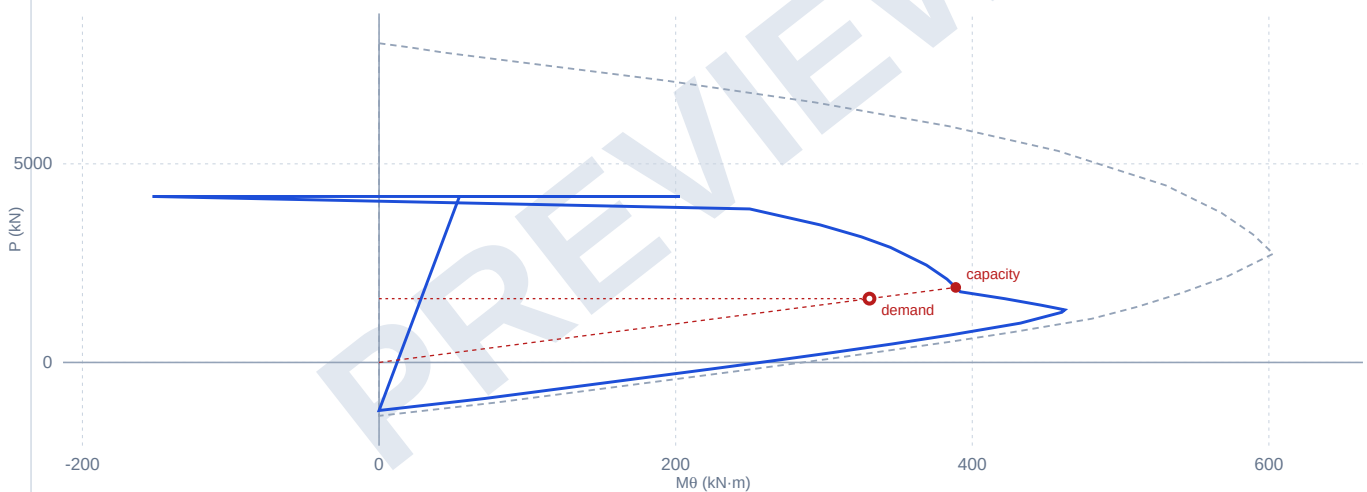


Bar fill: blue = compression, red = tension, grey = unstressed. The elevation links each bar to its own strain.

Exact direct meridian at the equilibrium strain direction

Direct $P-M\beta$ at $\beta_{eq} = 0.000^\circ$, with the load point

solid = adaptive Design · dashed = fixed Nominal · points are projected on β_{eq}



Concrete and resultants

Block area $A_{c,blk}$	84,071.4 mm ²	Block centroid	(0.0, 144.9) mm
Block stress σ_{block}	34.000 MPa	Concrete force C_c	2,858.4 kN
Concrete M_{cx}	414.2 kN·m	Concrete M_{cy}	0.0 kN·m
Nominal P_n	2,907.1 kN	Nominal M_{nx}	598.5 kN·m
Nominal M_{ny}	-0.0 kN·m	ϕ	0.6500
Design ϕP_n	1,889.6 kN	Design ϕM_{nx}	389.0 kN·m
Design ϕM_{ny}	-0.0 kN·m		

Reinforcement ledger at the solved state

Bar	X (mm)	Y (mm)	As (mm ²)	depth (mm)	ϵ	σ_s (MPa)	F (kN)	in block
1	-150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
2	0.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
3	150.0	-200.0	400.0	450.0	-1.909e-3	-381.82	-152.73	no
4	150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Bar	X (mm)	Y (mm)	As (mm²)	depth (mm)	ε	σs (MPa)	F (kN)	in block
5	150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
6	0.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
7	-150.0	200.0	400.0	50.0	2.455e-3	420.00	154.40	yes
8	-150.0	0.0	400.0	250.0	2.727e-4	54.55	21.82	no

Solver and admissibility evidence

Neutral-axis angle θ	90.0000°	Neutral-axis depth c	275.000 mm
Block depth a	210.179 mm	β1	0.7643
Projected depth Dθ	500.00 mm	Controlling εt	0.001909
Controlling bar	1	Classification	compression-controlled
Axial cap governs	no	Component residual	1.31e-10
Strain admissibility	admissible	Governing utilization	0.8500